

ODD Protocol for State-Dependent Participation Diffusion on Adaptive Influence Networks

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Abstract

This document describes the agent-based model in *State-Dependent Participation Diffusion on Adaptive Influence Networks* using the ODD protocol. The model represents recurrent collective participation on a weighted, directed, adaptive influence network. Agents may participate, broadcast, become inactive, and later return. Participation and social interaction change motivation, commitment, fatigue, and influence relationships, while external events, capacity, and temporary scaffolding alter the conditions under which agents participate, leave, or return. The implementation supports stochastic, deterministic expected-value, and ensemble simulation. The specification records the update sequence, initialization rules, parameter values, and sub-models used in the accompanying implementation.

1 Overview

1.1 Purpose and Patterns

The model studies recurrent collective participation on an adaptive influence network. Unlike one-time adoption models, it follows repeated participation, temporary inactivity, and return as internal states and influence relationships change. Voluntary student activities provide the motivating example.

The central question is whether temporary activation develops into sustained participation or disappears after external stimulation is withdrawn. The main patterns of interest are collapse after a temporary spike and continued high participation after external stimulation ends. The model also tracks threshold sensitivity and path dependence generated by accumulated commitment. Turnover and reactivation provide secondary population-level outcomes, while repeated co-participation and broadcasting change the influence network.

The reference experiments compare collapse-prone, persistent, and intervention-supported regimes. The control analysis varies event pulses, visibility, capacity, and temporary scaffolding.

1.2 Entities, State Variables, and Scales

The model contains n agents indexed by $i \in V$ and a weighted, directed influence network:

$$G_t = (V, E, W_t).$$

A directed edge $j \rightarrow i$ represents a channel through which agent j may influence agent i . The effective weight of this channel is

$$\omega_{ij}(t) = \text{clip}_{[0,1]}[v_{ij}(t)s_{ij}(t)],$$

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where $v_{ij} \in [0, 1]$ denotes visibility and $s_{ij}(t) \in [0, 1]$ denotes persuasiveness. The node set remains fixed during each simulation run. However, activity status and influence weights may change over time.

Each agent carries three core latent states: motivation $m_i(t) \in [0, 1]$, commitment $L_i(t) \in [0, 1]$, and fatigue $F_i(t) \in [0, 1]$. Optional modulators include affect $E_i(t) \in [0, 1]$, saturation $S_i(t) \in [0, 1]$, and stability gain $\Phi_i(t) \in [0, 1]$. Motivation is the fast state and governs immediate behavioral propensity. Commitment and fatigue change more slowly and affect later motivational decay. When enabled, the modulators adjust action propensity and the response to reinforcement, social load, and recent variation.

The activity variable $act_i(t)$ distinguishes active agents from those who are temporarily inactive. In stochastic simulations,

$$act_i(t) \in \{0, 1\},$$

whereas in deterministic expected-value simulations $act_i(t) \in [0, 1]$ is propagated as an expected active fraction. The two observable behavioral variables are participation $a_i(t)$ and broadcasting $p_i(t)$. In the motivating implementation, they correspond to attendance and posting. More generally, they may represent other costly and low-cost forms of recurrent action. Capacity adjustment converts intended participation $a_i(t)$ into realized participation $\tilde{a}_i(t)$. Aggregate realized participation is

$$A(t) = \sum_{i=1}^n \tilde{a}_i(t).$$

Agent-specific parameters govern action thresholds, decay, peer response, reinforcement, fatigue, dropout, and reactivation. The model also includes several externally supplied covariates. The availability factor $c_i(t) \in [0, 1]$ determines how much of an event input reaches agent i . The quantity $T_i(t) \geq 0$ represents the available engagement opportunity, while $\bar{y}_i > 0$ caps credited engagement. The role-salience variable $s_i^{\text{role}}(t)$ scales the commitment generated through socially reinforced participation. Finally, $r^+(t) \in [0, 1]$ represents favorable collective feedback. These quantities are scenario inputs or observable covariates, not latent states.

The population-level quantities are the viability threshold N_{min} , the run-level capacity K , the event schedule $\{(\tau_k, A_k)\}$, the run-level visibility multiplier γ_v , and the time-dependent scaffolding multiplier $\gamma_B(t)$. In the current implementation, K and γ_v remain constant within a simulation run. Event input and scaffolding, however, may vary across periods.

Time is discrete, with model periods

$$t = 0, 1, \dots, T - 1.$$

Displayed Week w corresponds to model period $t = w - 1$. Because actions are generated before the event input updates motivation, an event scheduled for Week w first affects participation in Week $w + 1$. For the 18-period experiments, the terminal four-week window is $t = 14, \dots, 17$. One model period represents one week in the accompanying implementation. Core latent states and normalized influence variables are projected onto $[0, 1]$ after each update.

1.3 Process Overview and Scheduling

Updates are synchronous: all period- t quantities are evaluated before any period- $t + 1$ state is written. Within-period outcomes depend only on the states and network available at the start of the period.

Each period proceeds from current states and network weights to actions, realized participation, state transitions, and network updating.

One period proceeds as follows:

1. read the period- t agent states, network weights, exogenous inputs, and controls;
2. calculate the motivation entering the action rule and the participation and broadcasting probabilities;
3. draw stochastic actions or substitute their deterministic expected values;
4. apply the capacity adjustment to obtain $\tilde{a}_i(t)$ and $A(t)$;
5. calculate peer-motivation input and weighted neighbor participation using the period- t network;
6. calculate external input, internal reinforcement, effective decay, dropout hazard, and reactivation probability;
7. update motivation, commitment, fatigue, optional modulators, and activity status synchronously;
8. update $v_{ij}(t+1)$ from period- t participation, broadcasting, and the persistent channel term;
9. calculate $cred_j(t+1)$, $s_{ij}(t+1)$, and $\omega_{ij}(t+1)$ for the next period;
10. record period- t actions and aggregate outputs.

The updated visibility, credibility, persuasiveness, and effective influence enter only the next period’s action rule. The snapshot labeled t contains the period- t states and influence weights together with the actions and aggregate participation generated from them. Values computed for $t+1$ first appear in snapshot $t+1$.

Stochastic, deterministic expected-value, and ensemble runs use this same transition order; only the treatment of action draws and randomized realizations differs.

2 Design Concepts

2.1 Basic principles

The model distinguishes one-time activation from recurrent participation. Participation is generated by a fast motivational state and modified by slower commitment and fatigue states. Realized participation feeds back into these states, changing both the probability and persistence of future behavior.

The influence structure is also endogenous. Visibility and persuasiveness change through public signaling, co-participation, commitment, and recent participation. Events, capacity, visibility, and scaffolding modify specified terms in the state transitions.

2.2 Emergence

Aggregate participation and its persistence or collapse emerge from individual state transitions and directed interaction. Turnover and intervention responses arise from the same process. Agents are not assigned fixed types such as “persistent” or “inactive.” Instead, these patterns arise from differences in thresholds, decay rates, reinforcement parameters, network positions, and prior state histories.

The main emergent contrast is between a temporary peak followed by collapse and a high-activity trajectory sustained after the initial stimulus. The piecewise reinforcement function creates sensitivity to the viability threshold. Path dependence and hysteresis can arise because commitment accumulated during earlier participation lowers later motivational decay

2.3 Adaptation

Agents adapt through changes in their internal states and influence relationships. Motivation responds quickly to peer input, external events, and realized reinforcement. Participation and collective feedback build commitment, while engagement and social exposure build fatigue; inactivity permits partial recovery.

The network adapts through visibility and credibility. Co-participation and broadcasting change exposure, while commitment and recent participation change persuasiveness.

2.4 Objectives

Individual agents have no explicit utility function or forward-looking objective. Their actions follow stochastic or deterministic threshold responses to current states.

The reverse planner is external to the agents. It searches over events, visibility, capacity, and temporary scaffolding to meet a terminal-window participation target at minimum control cost. The planner varies controls while leaving the transition equations unchanged.

2.5 Learning and Prediction

Agents do not learn strategically or estimate a model of their environment. Past experience affects behavior through accumulated state variables. Previous participation affects commitment and credibility, while repeated interaction changes visibility. These state variables provide memory without explicit belief updating or statistical learning.

Forward prediction occurs in the simulator and reverse planner, not within the agents. The planner evaluates each candidate control by propagating the model over the remaining horizon. The resulting projections are conditional on the selected parameters and initial state.

2.6 Sensing

Agent i 's action depends on its internal states, activity status, and exogenous inputs. Social information enters through weighted incoming influence: peer motivation is transmitted through $\omega_{ij}(t)$, and neighbor participation through $\bar{a}_{N(i)}(t)$.

Neighbors' motivation is not observed directly; it enters through the weighted peer signal in Equation (2). Event exposure depends on availability, event visibility, affect, and saturation.

2.7 Interaction

Interactions are directed and weighted. The influence of agent j on agent i depends on the product of visibility $v_{ij}(t)$ and persuasiveness $s_{ij}(t)$. Co-participation changes mutual visibility, while broadcasting changes the sender's directed visibility.

The network transmits motivational and behavioral signals, and resulting actions may later alter the network. Inactive agents do not participate or broadcast, but external input or active neighbors may reactivate them.

2.8 Stochasticity

Randomness enters through network generation, parameter and state initialization, action draws, shocks, dropout, and reactivation. In stochastic runs, participation and broadcasting are Bernoulli variables conditional on current action probabilities and activity status.

Deterministic expected-value runs replace these draws with conditional expectations and propagate expected activity status. The deterministic mode is a plug-in approximation: expected actions and activity fractions are inserted directly into the nonlinear transition equations. The reported deterministic ensembles still resample the network, heterogeneous parameters, and initial states.

2.9 Collectives

The network generator assigns agents to initial communities that create clustered ties and cross-community bridges. These communities shape exposure but have no separate states or objectives. Population-level conditions are represented by aggregate participation $A(t)$ and the viability threshold $N_{\min}$.

2.10 Observation

The primary observable outputs are individual participation, broadcasting, realized participation, activity status, and aggregate participation. The simulator also records motivation, commitment, fatigue, the optional modulator states, and effective influence weights. Motivational reach is the fraction of agents satisfying $m_i(t) > 0.4$.

Deterministic and ensemble analyses report means, terminal-window averages, and empirical quantiles across randomized realizations. Public-body dots are display elements outside the modeled node set V .

3 Details

3.1 Initialization

3.1.1 Network Initialization

Each simulation begins with n agents and a directed, clustered influence network. Let g denote the target friend-group size. The number of communities is

$$C = \max \left\{ 2, \text{round} \left(\frac{n}{g} \right) \right\}.$$

With $n = 36$ and $g = 8$, the reference network contains five communities of sizes 8, 7, 7, 7, and 7. Agents are assigned by index,

$$\text{comm}_i = i \bmod C.$$

Community labels are fixed by index, while angular positions and network ties vary with the random seed.

Agent i receives angular position

$$\theta_i = \frac{2\pi \text{comm}_i}{C} + U_i, \quad U_i \sim \text{Uniform} \left(-\frac{0.8\pi}{C}, \frac{0.8\pi}{C} \right),$$

and dyadic similarity is

$$\text{sim}_{ij} = \exp\left(-\frac{d_{ij}^2}{0.6}\right).$$

For each receiver i , the number of within-community incoming ties is

$$k_i^{\text{in}} = \text{round}(d_{\text{in}}) + U_i^{\text{in}}, \quad U_i^{\text{in}} \sim \text{DiscreteUniform}\{0, 1, 2\}.$$

With $d_{\text{in}} = 4$, each receiver forms incoming ties from the four to six most similar agents in its community. The number of cross-community bridge ties is

$$k_i^{\text{out}} = \lfloor d_{\text{out}} \rfloor + Z_i, \quad Z_i \sim \text{Bernoulli}(d_{\text{out}} - \lfloor d_{\text{out}} \rfloor).$$

With $d_{\text{out}} = 1.5$, each receiver obtains one or two uniformly sampled cross-community senders with equal probability. Self-loops and duplicate directed edges are prohibited. Reciprocal ties are allowed and generated independently.

For each standard within-community or bridge edge,

$$v_{ij}^{\text{mix}}(0) = \text{clip}_{[0,1]} [b_{ij} + 0.30 \text{sim}_{ij} + U_{ij}^v],$$

with b_{ij} and U_{ij}^v defined as above.

The implementation sets

$$\text{online}_{ij}(0) = \text{offline}_{ij}(0) = v_{ij}^{\text{mix}}(0), \quad \pi_o = \pi_f = 0.5,$$

and holds both channel components fixed. Hence,

$$v_{ij}^{\text{mix}}(t) = v_{ij}^{\text{mix}}(0) \quad \text{for all } t.$$

Initial adaptive visibility is

$$v_{ij}(0) = \text{clip}_{[0,1]} [0.5v_{ij}^{\text{mix}}(0) + 0.1].$$

Because no pre-simulation participation history is supplied,

$$\bar{a}_j^{\text{pre}} = 0, \quad \text{cred}_j(0) = 0.5L_j(0).$$

Initial persuasiveness and effective influence are

$$s_{ij}(0) = \text{clip}_{[0,1]} [0.30 + 0.40 \text{sim}_{ij} + 0.30 \text{cred}_j(0)], \quad \omega_{ij}(0) = \text{clip}_{[0,1]} [v_{ij}(0)s_{ij}(0)].$$

The master seed is $q_0 = 11$, and realization $r = 0, \dots, R - 1$ uses

$$q_r = 11 + 9973r.$$

Each realization resamples angular positions, directed ties, heterogeneous parameters, and initial states; scenario controls remain fixed.

3.1.2 Agent States and Parameters

I draw the initial motivation $m_i(0)$, commitment $L_i(0)$, and fatigue $F_i(0)$ from the regime-specific distributions in Table B2a and clip each draw to $[0, 1]$. When the optional modulators are enabled, I initialize

$$E_i(0) = \bar{E}_i, \quad S_i(0) = 0, \quad \Phi_i(0) = 1,$$

together with

$$\bar{m}_i(0) = m_i(0), \quad \text{var}_i(0) = 0.$$

All agents begin active:

$$\text{act}_i(0) = 1.$$

Agent-specific parameters govern action thresholds, response slopes, motivational decay, peer influence, reinforcement, fatigue, dropout, reactivation, visibility, and persuasiveness. Unless a quantity is defined as a dynamic state or time-dependent input, its sampled value remains fixed throughout the realization.

The reference experiments use two parameter regimes. The collapse-prone regime combines faster motivational decay, weaker reinforcement and peer transmission, and a higher viability threshold. The favorable regime uses slower decay, stronger reinforcement and peer transmission, and a lower viability threshold. These regimes are parameter configurations rather than fixed agent types.

Tables B1 to B3 report the parameter values, initialization distributions, and sampling status used in the implementation. A quantity marked "Fixed" remains constant across the stated regime. A quantity marked "Randomized once" is sampled during initialization and then held fixed within that realization. Quantities marked "Dynamic" are updated by the model equations or supplied as time-dependent inputs.

3.1.3 Controls and Reference Scenarios

I fix the external controls before each forward simulation unless I use a receding-horizon procedure. An event schedule specifies the event times τ_k and amplitudes A_k . In the current implementation, I hold the visibility multiplier γ_v and capacity K constant within each simulation run. Temporary scaffolding may vary across periods through $\gamma_B(t)$. Outside the scaffolding window, $\gamma_B(t) = 1$. The parameter s_B determines the additive below-quorum support supplied while scaffolding is active.

The computational experiments reported in the associated paper use $n = 36$ agents over an 18-period horizon. The model periods are $t = 0, \dots, 17$, which are displayed as Weeks 1–18. Displayed Week w corresponds to $t = w - 1$. Event and scaffolding labels in Table 1 follow the displayed-week convention. Three reference scenarios are compared:

Scenario	Regime	Events	Visibility	Capacity	Scaffolding
A	Collapse-prone	Week 3 amplitude 0.60	$\gamma_v = 1.0$	$K = 14$	None
B	Favorable	Weeks 3, 7, 11 amplitude 0.35 each	$\gamma_v = 1.4$	$K = 18$	None
C	Collapse-prone	Weeks 2, 5, 8, 12 amplitude 0.45 each	$\gamma_v = 3.0$	$K = 20$	Weeks 2–9 $\gamma_B = 1.4$

Table 1: Reference control configurations used in the computational experiments

Scenario C is the rescue configuration used in the 18-period experiment.

In the Figure 1 parameter sweep, I hold the Scenario A event schedule, capacity, and scaffolding settings fixed. I vary only N_{min} and γ_v .

Figures 1 and 2 use deterministic expected-value actions, with $a_i(t)$, $p_i(t)$, and activity status propagated through their conditional expectations.

In the reported reference experiments, I activate the affect, saturation, and stability modulators with the nonzero coupling values listed in Table B1. I set the idiosyncratic motivation shock to $\epsilon_i(t) = 0$ in the deterministic experiments. Each realization resamples the network, heterogeneous parameters, and initial states while retaining the same scenario controls, horizon, population size, and regime. Each cell in the viability-visibility sweep averages 20 independent realizations. The trajectory comparison averages $R = 250$ independent realizations for each reference scenario.

The realization seeds follow the master-seed rule in Table B3, and the computational materials record the seed sequence and implementation version used for the figures.

3.2 Input Data

The current implementation uses no empirical input data. I generate the networks, initial states, and heterogeneous parameters synthetically from the distributions of the selected regime. The user or a reference scenario supplies the event schedule, visibility amplification, capacity, scaffolding, simulation horizon, random seed, and ensemble size.

In synthetic runs, I generate the externally supplied agent-level quantities $c_i(t)$, $T_i(t)$, $s_i^{role}(t)$, and the population-level signal $r^+(t)$ from scenario settings. An empirical implementation would require these inputs to be estimated or constructed from longitudinal participation and network data.

3.3 Submodels

I use the following definitions throughout the submodels:

$$\sigma(x) = \frac{1}{1 + e^{-x}}, \quad \text{clip}_{[0,1]}(x) = \min\{1, \max\{0, x\}\}, \quad [x]_+ = \max\{x, 0\}, \quad [x]_- = \max\{-x, 0\},$$

and

$$\mathbf{1}\{\mathcal{C}\} = \begin{cases} 1, & \text{if condition } \mathcal{C} \text{ is true,} \\ 0, & \text{otherwise.} \end{cases}$$

The constant $\epsilon_0 > 0$ prevents division by zero. The term $\epsilon_i(t)$ is a zero-mean idiosyncratic motivation shock with the distribution reported in Table B2; In the deterministic experiments, however, I set $\epsilon_i(t) = 0$.

3.3.1 Actions and Realized Participation

Emotion-shifted motivation is

$$\hat{m}_i(t) = \text{clip}_{[0,1]} \left[m_i(t) + \beta_E \left(E_i(t) - \frac{1}{2} \right) \right]$$

Participation and broadcasting probabilities are

$$P_i^a(t) = \sigma(\kappa_i^a[\hat{m}_i(t) - \theta_i^a]),$$

$$P_i^p(t) = \sigma(\kappa_i^p[\hat{m}_i(t) - \theta_i^p]). \quad (1)$$

In stochastic simulations,

$$Z_i^a(t) \sim \text{Bernoulli}(P_i^a(t)), \quad Z_i^p(t) \sim \text{Bernoulli}(P_i^p(t)),$$

and

$$a_i(t) = \text{act}_i(t)Z_i^a(t), \quad p_i(t) = \text{act}_i(t)Z_i^p(t).$$

In deterministic expected-value simulations,

$$a_i(t) = \text{act}_i(t)P_i^a(t), \quad p_i(t) = \text{act}_i(t)P_i^p(t).$$

I convert intended participation into realized participation through the capacity constraint:

$$\tilde{a}_i(t) = \frac{a_i(t)}{1 + \frac{\sum_j a_j(t)}{K}} = a_i(t) \frac{K}{K + \sum_j a_j(t)}. \quad (14)$$

Aggregate realized participation is:

$$A(t) = \sum_{i=1}^n \tilde{a}_i(t).$$

3.3.2 Peer Input, Motivation, and Decay

I define the weighted local participation aggregate as:

$$\bar{a}_{N(i)}(t) = \frac{\sum_j \omega_{ij}(t) \tilde{a}_j(t)}{\sum_j \omega_{ij}(t) + \epsilon_0}.$$

The peer-motivation signal received by agent i is

$$X_i(t) = \sum_j \omega_{ij}(t) b_j \tanh(\kappa_g m_j(t)),$$

$$\tilde{X}_i(t) = \frac{X_i(t)}{1 + \rho_i X_i(t)}. \quad (2)$$

I update motivation according to

$$m_i(t+1) = \text{clip}_{[0,1]} \left[(1 - \alpha_i(t)) m_i(t) + G(\Phi_i(t)) (w_i^P (1 - S_i(t)) \tilde{X}_i(t) + c_i(t) R_i(t) + B_i(t) + \epsilon_i(t)) \right] \quad (3)$$

where

$$G(\Phi) = 1 + \kappa_\Phi (1 - 2\Phi), \quad \kappa_\Phi \in [0, 1).$$

The effective fatigue entering the decay equation is:

$$\hat{F}_i(t) = \text{clip}_{[0,1]} \left[F_i(t) + \kappa_{EF} \left(\frac{1}{2} - E_i(t) \right) \right].$$

The endogenous motivational decay rate is then:

$$\alpha_i(t) = \text{clip}_{[0,1]} \left[\alpha_{0,i} (1 - \eta_L [1 - \kappa_{SL} S_i(t)] L_i(t)) (1 + \eta_F \widehat{F}_i(t)) \right]. \quad (4)$$

3.3.3 Commitment, Fatigue, and Affective Modulation

I use separate multipliers to represent the effects of affect on fatigue and positive reinforcement:

$$\nu_F(E) = 1 + \beta_F \left(\frac{1}{2} - E \right),$$

$$\nu_R(E) = 1 - \beta_R \left(\frac{1}{2} - E \right),$$

with

$$0 \leq \beta_F, \beta_R \leq 2.$$

I update commitment as:

$$\begin{aligned} L_i(t+1) = \text{clip}_{[0,1]} \Big[& L_i(t) + \lambda_L (\phi_1 \tilde{a}_i(t) + \phi_2 s_i^{\text{role}}(t) (1 - \kappa_{SL} S_i(t)) \bar{a}_{N(i)}(t) \\ & + \phi_3 \nu_R(E_i(t)) r^+(t) - L_i(t)) \Big]. \end{aligned} \quad (5)$$

I update fatigue as:

$$\begin{aligned} F_i(t+1) = \text{clip}_{[0,1]} \Big[& F_i(t) + \lambda_F (\nu_F(E_i(t)) \psi_1 y_i(t) + \nu_F(E_i(t)) \psi_2 \bar{a}_{N(i)}(t) \\ & + \psi_3 (1 - \tilde{a}_i(t)) - F_i(t)) \Big]. \end{aligned} \quad (6)$$

3.3.4 External and Internal Reinforcement

I represent external stimulation as a sequence of exponentially decaying event pulses:

$$R_i(t) = \kappa_R [1 - S_i(t)] \nu_R(E_i(t)) V_i^{ev} \sum_{k: \tau_k \leq t} A_k e^{-\lambda_R (t - \tau_k)}, \quad \kappa_R = 3.4. \quad (7)$$

Internal reinforcement follows the piecewise function:

$$B_i(t) = \nu_R(E_i(t)) \begin{cases} \gamma_B(t) \beta_i (1 - e^{-\delta_i y_i(t)}) - \chi_i \mathbf{1}_{\{y_i(t)=0\}}, & A(t) \geq N_{\min}, \\ -\chi_i^{\text{viab}} + [\gamma_B(t) - 1] s_B, & A(t) < N_{\min}. \end{cases} \quad (8)$$

The corresponding reinforcement half-saturation point is

$$y_{i,1/2} = \frac{\ln 2}{\delta_i}.$$

Above quorum, scaffolding amplifies productive reinforcement. Below quorum, it offsets part of the viability penalty without amplification of the nonparticipation penalty.

3.3.5 Adaptive Influence Weights

I define effective influence as:

$$\omega_{ij}(t) = \text{clip}_{[0,1]}(v_{ij}(t)s_{ij}(t)). \quad (9)$$

The persistent channel mixture is defined as:

$$v_{ij}^{mix}(t) = \pi_{online}e_{ij}(t) + \pi_{offline}e_{ij}(t), \quad \pi_o + \pi_f = 1. \quad (12)$$

I calculate persuasiveness as:

$$s_{ij}(t) = \text{clip}_{[0,1]} [\zeta_i^0 + \zeta_i^1 \text{sim}_{ij} + \zeta_i^2 \text{cred}_j(t)], \quad (10)$$

where

$$\text{cred}_j(t) = w_1^c L_j(t) + w_2^c H_j(t), \quad (11)$$

$$d_t = \min\{\Delta, t\},$$

and

$$H_j(t) = \begin{cases} 0, & t = 0, \\ \frac{1}{d_t} \sum_{\tau=t-d_t}^{t-1} \tilde{a}_j(\tau) & t \geq 1. \end{cases}$$

Credibility at time t uses $L_j(t)$ and at most the the preceding Δ realized-participation observations. After the period- t states and participation history are updated, Equations (10)–(11) determine $s_{ij}(t+1)$ and $\omega_{ij}(t+1)$ for the next behavioral update.

The visibility multiplier acts through the sender's effective expressive reach:

$$V_{j,\text{eff}}^{ex}(t) = \text{clip}_{[0,1]} [\gamma_v V_{j,0}^{ex}].$$

I then update visibility as:

$$v_{ij}(t+1) = \text{clip}_{[0,1]} \left[(1 - \mu)v_{ij}(t) + \mu(\eta_1 \tilde{a}_i(t) \tilde{a}_j(t) + \eta_2 V_{j,\text{eff}}^{ex} p_j(t)) + \mu_{\text{mix}}(v_{ij}^{\text{mix}}(t) - v_{ij}(t)) \right]. \quad (13)$$

The visibility control changes the opportunity for exposure, but persuasiveness remains unmodified.

3.3.6 Attrition and Reactivation

I define the dropout hazard as

$$h_i(t) = h_{0,i} \exp \left[-\gamma_1 m_i(t) - \gamma_2 L_i(t) + \gamma_3 F_i(t) \right]. \quad (15)$$

Conditional on agent i being active,

$$d_i(t) \sim \text{Bernoulli}(1 - e^{-h_i(t)}).$$

For an inactive agent, the reactivation probability is:

$$P_i^{\text{react}}(t) = 1 - \exp \left[-\xi_1 R_i(t) - \xi_2 \bar{a}_{N(i)}(t) \right].$$

$$r_i(t) \sim \text{Bernoulli}(P_i^{\text{react}}(t)). \quad (16)$$

In stochastic simulations, I update activity status as:

$$act_i(t+1) = act_i(t)[1 - d_i(t)] + [1 - act_i(t)]r_i(t).$$

In deterministic expected-value simulations, I use:

$$act_i(t+1) = act_i(t)e^{-h_i(t)} + [1 - act_i(t)]P_i^{react}(t).$$

3.3.7 Optional Modulators

I update affect according to:

$$E_i(t+1) = \text{clip}_{[0,1]} \left[(1 - \lambda_E)E_i(t) + \lambda_E \bar{E}_i + \zeta_E[B_i(t)]_+ - \zeta'_E[B_i(t)]_- + \kappa_E \xi_i^E(t) \right]. \quad (17)$$

I define the incoming social load as:

$$\ell_i(t) = \sum_j \omega_{ij}(t), \quad (18a)$$

I then update saturation as:

$$S_i(t+1) = act_i(t) \left[(1 - \lambda_S)S_i(t) + \lambda_S \frac{\ell_i(t)}{\ell_i(t) + s_0} \right]. \quad (18b)$$

I calculate the moving mean and variance of motivation as:

$$\bar{m}_i(t) = (1 - \lambda_v)\bar{m}_i(t-1) + \lambda_v m_i(t), \quad (19a)$$

$$\text{var}_i(t) = (1 - \lambda_v)\text{var}_i(t-1) + \lambda_v (m_i(t) - \bar{m}_i(t))^2, \quad (19b)$$

The resulting stability gain is

$$\Phi_i(t) = \frac{1}{1 + \kappa_{\text{var}}\text{var}_i(t)}. \quad (19c)$$

3.3.8 Reverse Planning and Ensemble Evaluation

I use u to represent event timing and amplitudes, the run-level visibility multiplier γ_v , the run-level capacity K , and the time-dependent scaffolding schedule $\gamma_B(t)$. In the current implementation, γ_v and K are constant within each candidate simulation run. The planner searches for a low-cost control that satisfies the terminal-window target:

$$\bar{A}_{T-w:T-1}(u) = \frac{1}{w} \sum_{t=T-w}^{T-1} A(t) \geq A^*. \quad (20)$$

A representative additive cost is

$$C(u) = \kappa_{E,0}N_E + \kappa_{E,1} \sum_{k=1}^{N_E} A_k + \kappa_v(\gamma_v - 1) + \kappa_B \sum_t [\gamma_B(t) - 1] + \kappa_K[K - K_0]_+.$$

Here, N_E is the number of scheduled events, and K_0 is the uncontrolled baseline capacity. Table B1 reports the nonnegative control-cost coefficients $\kappa_{E,0}$, $\kappa_{E,1}$, κ_B , and κ_K . Because γ_v and K are run-level controls, their costs enter only once per simulation run. Scaffolding, however, is time dependent. Its cost is therefore summed across the periods in which it is active.

I use a bounded multi-start heuristic. Let m denote the maximum permitted number of events. The heuristic begins by constructing three event-timing templates. The first template places maximum-amplitude events at displayed weeks

$$w_k = 1 + \text{round}\left(\frac{k \min\{T-2, 12\}}{\max\{1, m\}}\right), \quad k = 0, \dots, m-1.$$

The second template distributes the same number of maximum-amplitude events evenly from the displayed-week $0.10(T-1)$ to $0.70(T-1)$. The third uses the interval from $0.25(T-1)$ to $0.80(T-1)$. Event weeks are rounded and clipped to the admissible range $1, \dots, T-1$.

I combine each event-timing template with a bounded set of capacity candidates. Let $K_{\max}$ denote the user-specified capacity ceiling. The capacity grid begins at $\min\{K_0, K_{\max}\}$ and increases in increments of two. I also include $K_{\max}$ when it does not fall exactly on this grid. Under the default values $K_0 = 14$ and $K_{\max} = 20$, the tested capacities are

$$K \in \{14, 16, 18, 20\}.$$

All starting configurations use the permitted visibility ceiling. When scaffolding is allowed, I set $\gamma_B = 2.0$ over displayed Weeks 1-6 and truncate this window when the simulation horizon is shorter. When scaffolding is not allowed, I set $\gamma_B(t) = 1$ throughout the candidate run.

I first evaluate the resulting starting configurations on the deterministic reference realization. If none satisfies the terminal target, the procedure reports that no feasible plan was found. It also displays the starting configuration with the highest terminal-window participation.

I order feasible starting configurations by control cost, from lowest to highest. If two configurations have the same cost, I evaluate the one with higher terminal-window participation first. Subject to the remaining global evaluation budget, I apply greedy control reduction to the feasible starts in this order. The global budget may be exhausted before every feasible start receives a complete descent. The procedure retains the lowest-cost feasible result found among the search paths that it actually evaluates.

When no immediate cost-reducing move remains feasible, the search tests one-period shifts in the existing event times. If enough of the evaluation budget remains, it may also evaluate up to five seeded perturbation starts. A perturbation may strengthen existing controls or add an event at another admissible week before greedy descent is resumed. Scenario C in Table 1 is a hand-specified rescue benchmark rather than an output of the heuristic planner.

Candidate controls are evaluated on the deterministic reference realization with seed 11. The planner enforces the terminal-window target without a downward feasibility tolerance. One global budget of at most 400 candidate-control evaluations covers the starting configurations, feasibility checks during greedy descent, event-timing changes, and perturbation searches. Each individual descent is limited to at most 60 passes. Therefore, some feasible starts may not receive a complete descent, and the procedure may evaluate fewer than five perturbation starts. The perturbation generator uses seed 12. Event amplitudes are reduced in increments of 0.05, visibility in increments of 0.25, scaffolding

strength in increments of 0.15, and scaffolding duration in one-period increments. Capacity candidates are separated by increments of two agents, with the user-specified ceiling included. When $\kappa_K > 0$, the descent procedure may also reduce capacity by two agents. It accepts this change only if the stated cost decreases and feasibility is preserved. Under the reported setting $\kappa_K = 0$, the control cost is unaffected by capacity, but the different capacity values remain distinct feasibility candidates.

The default reverse-planning interface uses $n = 36$, $T = 18$, the collapse-prone regime, terminal target $A^* = 9$, a maximum of four events, an event-amplitude ceiling of 0.60, a visibility ceiling of $\gamma_v = 4.0$, a capacity ceiling of $K = 20$, and permission to use temporary scaffolding. The default interface searches within these bounds; changing them may alter both feasibility and the plan returned by the heuristic.

I run the planner on one fixed deterministic reference realization. I then evaluate robustness separately over randomized ensembles. For scalar output $Z(t)$, the ensemble mean is

$$\bar{Z}_R(t) = \frac{1}{R} \sum_{r=1}^R Z^{(r)}(t).$$

Empirical quantiles summarize the variation across network, initial-state, and parameter realizations. A failed search indicates only that no feasible schedule was found within the evaluated candidates.

A Visualization and Interface Specification

A.1 Motivation Color Map

I use node fill color to represent motivation $m_i(t)$. The display uses a fixed plasma color map defined by six anchor colors placed at levels $\ell_k = k/5$.

k	Level	R	G	B
0	0.0	13	8	135
1	0.2	126	3	168
2	0.4	177	42	144
3	0.6	225	100	98
4	0.8	252	166	54
5	1.0	240	249	33

Table 2: Plasma anchor colors used to represent motivation in the simulator

For $m \in [0, 1]$, let $u = 5m$, $k = \lfloor u \rfloor$ capped at 4, and $f = u - k$. I calculate the rendered color as:

$$C(m) = (1 - f)c_k + fc_{k+1}.$$

I hold this mapping fixed across all agents, periods, scenarios, and simulation runs.

A.2 Core Nodes, Commitment Arcs, and Contextual Public Nodes

An outer ring identifies each modeled agent. The function $C(m_i(t))$ determines the node fill, while an arc with angular extent $2\pi L_i(t)$ represents commitment.

Inactive modeled agents are rendered with a desaturated or dashed style.

Public-body dots are contextual display elements outside the modeled node set V . They may be animated to illustrate reach but are excluded from all state updates and reported model statistics.

A.3 Aggregate Readouts

Mean motivation is

$$\bar{m}(t) = \frac{1}{n} \sum_{i=1}^n m_i(t).$$

For threshold τ , motivational reach is

$$\rho_\tau(t) = \frac{1}{n} |\{i : m_i(t) > \tau\}|.$$

Cross-sectional motivational dispersion is

$$s_m(t) = \sqrt{\frac{1}{n} \sum_{i=1}^n [m_i(t) - \bar{m}(t)]^2}$$

I also report realized participation, expected active population, mean commitment, mean fatigue, motivational reach, terminal-window participation, and quorum-attainment or terminal-target-attainment rates.

A.4 Ensemble Display

Ensemble trajectories display the mean and, when available, empirical quantile bands across randomized realizations. For a scalar or indexed node quantity $Z_i(t)$,

$$\bar{Z}_i(t) = \frac{1}{R} \sum_{r=1}^R Z_i^{(r)}(t).$$

Node-level ensemble averages refer to indexed simulated agents or matched structural positions.

In ensemble display mode, node states are shown as ensemble averages, while animated edges come from the realization whose terminal-window outcome is closest to the ensemble mean.

B Parameter, Initialization, and Reproducibility Tables

Tables B1–B3 report the values used in the current implementation. “Fixed” means that the quantity remains constant across agents and realizations within the stated regime. “Randomized once” means that the quantity is sampled during initialization and then held fixed throughout that realization. “Dynamic” means that the model equations update the quantity over time. For $X \sim \mathcal{N}(\mu, \sigma^2)$, I define the clipped-normal draw as:

$$\text{CN}_{[a,b]}(\mu, \sigma) = \min\{b, \max\{a, X\}\}.$$

The implementation draws one value from the stated normal distribution and then clips it to the stated interval. It does not redraw a value when the original draw falls outside that interval.

Table B1. Global, Dynamic, and Control Parameters

Symbol	Meaning	Allowed range	Collapse-prone setting	Favorable setting	Status
n	Number of modeled agents	$\mathbb{Z}^+$	36	36	Fixed
T	Number of simulated periods	$\mathbb{Z}^+$	18	18	Fixed
N_{min}	Minimum viable level of realized participation	$0, \dots, n$	10; sweep 2 – 12	4	Fixed within run
K	Participation capacity	> 0	$A : 14$; $C : 20$	$B : 18$	Fixed within run
γ_v	Run-level visibility multiplier	≥ 1	$A : 1.0$; $C : 3.0$; sweep 1.0-4.0 by 0.5	$B : 1.4$	Fixed within run
$\gamma_B(t)$	Time-dependent scaffolding multiplier	≥ 1	$A : 1$; $C : 1.4$ during window	1	Dynamic input
s_B	Below-quorum scaffolding support	≥ 0	0.62	0.62	Fixed
λ_R	Event-pulse decay rate	> 0	0.60	0.60	Fixed
κ_R	Event-input gain	> 0	3.4	3.4	Fixed
$\{(\tau_k, A_k)\}$	Event schedule	$A_k \geq 0$	Scenario specific	Scenario specific	Input
ϵ_0	Safeguard against division by zero	> 0	10^{-6}	10^{-6}	Fixed
R	Number of ensemble realizations	$\mathbb{Z}^+$	250 for trajectory ensembles; 20 per viability-visibility cell	250 for trajectory ensembles; 20 per viability-visibility cell	Fixed
w	Length of the terminal averaging window	$\mathbb{Z}^+$	4	4	Fixed
A^*	Terminal participation target	$[0, n]$	9	9	Fixed
Δ	Credibility-history window	$\mathbb{Z}^+$	3	3	Fixed
π_o, π_f	Online/offline channel weights	$[0, 1]$, sum to 1	0.5, 0.5	0.5, 0.5	Fixed
μ	Activity-dependent visibility update rate	$[0, 1]$	0.28	0.28	Fixed
μ_{mix}	Persistent-channel update rate	$[0, 1]$	0.12	0.12	Fixed
η_1, η_2	Co-participation and broadcasting effects	≥ 0	0.55, 0.45	0.55, 0.45	Fixed
w_1^c, w_2^c	Credibility weights	≥ 0	0.5, 0.5	0.5, 0.5	Fixed

Table B1. Global, Dynamic, and Control Parameters

Symbol	Meaning	Allowed range	Collapse-prone setting	Favorable setting	Status
κ_g	Sender-signal saturation slope	> 0	2.0	2.0	Fixed
η_L	Commitment protection of motivation	≥ 0	0.93	0.62	Fixed
η_F	Fatigue effect on motivation decay	≥ 0	0.55	0.35	Fixed
κ_{SL}	Saturation weakening of commitment	$[0, 1]$	0.50	0.50	Fixed
λ_L	Commitment adjustment rate	$[0, 1]$	0.37	0.37	Fixed
λ_F	Fatigue adjustment rate	$[0, 1]$	0.30	0.30	Fixed
ϕ_1, ϕ_2, ϕ_3	Commitment target weights	≥ 0	0.68, 0.34, 0.20	0.68, 0.34, 0.20	Fixed
ψ_1, ψ_2, ψ_3	Fatigue target weights	≥ 0	0.45, 0.20, 0.30	0.45, 0.20, 0.30	Fixed
$\gamma_1, \gamma_2, \gamma_3$	Dropout-hazard state coefficients	≥ 0	1.6, 1.6, 1.3	1.6, 1.1, 1.2	Fixed
ξ_1, ξ_2	Reactivation coefficients	≥ 0	1.4, 1.5	1.5, 1.1	Fixed
$\beta_E, \beta_F, \beta_R$	Affect couplings	Stated model bounds	0.15, 0.30, 0.30	0.15, 0.30, 0.30	Fixed
κ_Φ	Stability-response coupling	$[0, 1)$	0.3	0.3	Fixed
$\lambda_E, \lambda_S, \lambda_v$	Optional-state update rates	$[0, 1]$	0.15, 0.10, 0.20	0.15, 0.10, 0.20	Fixed
κ_{var}	Variance-to-stability coefficient	≥ 0	18	18	Fixed
K_0	Baseline capacity in cost function	$[0, n]$	14	14	Fixed
$\kappa_{E,0}, \kappa_{E,1}$	Event cost coefficients	≥ 0	1.0, 2.2	1.0, 2.2	Fixed
$\kappa_v, \kappa_B, \kappa_K$	Visibility, scaffolding, and capacity costs	≥ 0	0.9, 1.4, 0.0	0.9, 1.4, 0.0	Fixed
κ_{EF}	Affect adjustment to effective fatigue	≥ 0	0.30	0.30	Fixed
ζ_E, ζ'_E	Positive and negative reinforcement effects on affect	≥ 0	0.30, 0.30	0.30, 0.30	Fixed

Table B2a. Agent-Level Values: Core Initial States

Symbol	Meaning	Range	Collapse-prone distribution	Favorable distribution	Status
$m_i(0)$	Initial motivation	$[0, 1]$	$\text{CN}_{[0,1]}(0.28, 0.10)$	$\text{CN}_{[0,1]}(0.33, 0.10)$	Randomized once
$L_i(0)$	Initial commitment	$[0, 1]$	$\text{CN}_{[0,1]}(0.10, 0.05)$	$\text{CN}_{[0,1]}(0.12, 0.06)$	Randomized once
$F_i(0)$	Initial fatigue	$[0, 1]$	$\text{CN}_{[0,1]}(0.35, 0.10)$	$\text{CN}_{[0,1]}(0.28, 0.10)$	Randomized once
$\bar{E}_i$	Affect baseline	$[0, 1]$	$\text{CN}_{[0,1]}(0.45, 0.12)$	$\text{CN}_{[0,1]}(0.22, 0.07)$	Randomized once
$E_i(0)$	Initial affect	$[0, 1]$	$E_i(0) = \bar{E}_i$	$E_i(0) = \bar{E}_i$	Initialized as $E_i(0) = \bar{E}_i$
$S_i(0)$	Initial saturation	$[0, 1]$	0	0	Fixed
$\Phi_i(0)$	Initial stability gain	$[0, 1]$	1	1	Fixed
$act_i(0)$	Initial activity status	$\{0,1\}$ or $[0, 1]$	1	1	Fixed
$\bar{m}_i(0)$	Initial moving mean	$[0,1]$	$\bar{m}_i(0) = m_i(0)$	$\bar{m}_i(0) = m_i(0)$	Initialized
$\text{var}_i(0)$	Initial moving variance	$[0, 1]$	0	0	Fixed

Table B2b. Agent-Level Values: Behavioral Parameters

Symbol	Meaning	Range	Collapse-prone distribution	Favorable distribution	Status
θ_i^a	Participation threshold	$[0, 1]$	$\text{CN}_{[0.1,0.95]}(0.60, 0.08)$	$\text{CN}_{[0.1,0.95]}(0.56, 0.10)$	Randomized once
θ_i^p	Broadcasting threshold	$[0, 1]$	$\text{CN}_{[0.1,0.98]}(0.70, 0.08)$	$\text{CN}_{[0.1,0.98]}(0.70, 0.12)$	Randomized once
κ_i^a	Participation-response slope	> 0	9	8	Fixed
κ_i^p	Broadcasting-response slope	> 0	11	9	Fixed
$\alpha_{0,i}$	Baseline motivation decay	$[0, 1]$	$\text{CN}_{[0.02,0.95]}(0.45, 0.10)$	$\text{CN}_{[0.02,0.95]}(0.28, 0.08)$	Randomized once
w_i^P	Peer-input susceptibility	≥ 0	$\text{CN}_{[0,1.2]}(0.28, 0.10)$	$\text{CN}_{[0,1.2]}(0.40, 0.15)$	Randomized once
b_i	Sender spreading propensity	≥ 0	$\text{CN}_{[0,1]}(0.30, 0.10)$	$\text{CN}_{[0,1]}(0.46, 0.10)$	Randomized once
ρ_i	Peer-input saturation coefficient	≥ 0	$\text{CN}_{[0.3,3]}(1.30, 0.20)$	$\text{CN}_{[0.3,3]}(1.50, 0.25)$	Randomized once

Table B2b. Agent-Level Values: Behavioral Parameters

Symbol	Meaning	Range	Collapse-prone distribution	Favorable distribution	Status
β_i	Maximum productive reinforcement	≥ 0	$\text{CN}_{[0,3]}(0.26, 0.08)$	$\text{CN}_{[0,3]}(0.34, 0.11)$	Randomized once
δ_i	Reinforcement saturation rate	> 0	2.0	1.25	Fixed
χ_i	Nonparticipation penalty	≥ 0	$\text{CN}_{[0,0.5]}(0.06, 0.03)$	$\text{CN}_{[0,0.5]}(0.05, 0.03)$	Randomized once
χ_i^{viab}	Below-quorum penalty	≥ 0	0.016	0.040	Fixed
$h_{0,i}$	Baseline dropout hazard	$[0, 0.3]$	$\text{CN}_{[0,0.3]}(0.06, 0.02)$	$\text{CN}_{[0,0.3]}(0.05, 0.02)$	Randomized once
V_i^{ev}	Event visibility	$[0, 1]$	0.70	0.72	Fixed
$V_{i,0}^{\text{ex}}$	Baseline expressive reach	$[0, 1]$	$\text{CN}_{[0,1]}(0.25, 0.10)$	$\text{CN}_{[0,1]}(0.60, 0.13)$	Randomized once
ζ_i^0	Baseline trust	$[0, 1]$	0.3	0.3	Fixed
ζ_i^1	Similarity weight	≥ 0	0.40	0.40	Fixed
ζ_i^2	Credibility weight	≥ 0	0.30	0.30	Fixed
s_0	Global saturation half-load parameter	> 0	2.0	2.0	Fixed

Table B2c. Agent-Level Values: Inputs and Shocks

Symbol	Exact Implementation
$c_i(t)$	Drawn once as $\text{CN}_{[0.1,1]}(0.55, 0.12)$, then constant over time
$T_i(t)$	1 for every agent and period
$\bar{y}_i$	1
$s_i^{\text{role}}(t)$	1
$r^+(t)$	0.10
$\epsilon_i(t)$	0 in deterministic mode; Uniform($-0.01, 0.01$) in stochastic mode
$\kappa_E \xi_i^E(t)$	0 in deterministic mode; Uniform($-0.05, 0.05$) in stochastic mode

Table B3. Network Generation, Initialization, and Reproducibility Settings

Component	Exact Implementation
Number of communities C	$C = \max\{2, \text{round}(n/g)\}$; reference $C = 5$
Community-size rule	For $n = 36$, $g = 8$, the community sizes are 8, 7, 7, 7, 7

Table B3. Network Generation, Initialization, and Reproducibility Settings

Component	Exact Implementation
Community assignment	$\text{comm}_i = i \bmod C$
Angular position	Community-sector center with an added draw from $\text{Uniform}(-0.8\pi/C, 0.8\pi/C)$
Within-community tie rule	Each receiver selects $4 + U$ most similar senders from the same community. $U \sim \text{DiscreteUniform}\{0, 1, 2\}$
Cross-community bridge rule	Each receiver selects one or two uniformly sampled senders from other communities, with probabilities 0.5, 0.5
Edge direction	A selected sender j forms directed edge $j \rightarrow i$
Self-loops	Prohibited
Reciprocal ties	Allowed but generated independently
sim_{ij}	$\exp(-d_{ij}^2/0.6)$
Duplicate directed edges	Prohibited
$\text{online}_{ij}(0)$	Equal to $v_{ij}^{\text{mix}}(0)$
$\text{offline}_{ij}(0)$	Equal to $v_{ij}^{\text{mix}}(0)$
π_o, π_f	0.5, 0.5
$v_{ij}(0)$	$\text{clip}_{[0,1]}[0.5v_{ij}^{\text{mix}}(0) + 0.1]$
$\bar{a}_j^{\text{pre}}$	0
$\text{cred}_j(0)$	$0.5L_j(0)$
$s_{ij}(0)$	$\text{clip}_{[0,1]}[0.30 + 0.40\text{sim}_{ij} + 0.30\text{cred}_j(0)]$
$\omega_{ij}(0)$	$\text{clip}_{[0,1]}[v_{ij}(0)s_{ij}(0)]$
Additional acquaintance ties	0 in all reported reference experiments
Network structure	Resampled for each realization
Heterogeneous parameters	Resampled for each realization
Initial latent states	Resampled for each realization
Master seed	11
Ensemble seed sequence	$q_r = 11 + 9973r, r = 0, \dots, R - 1$
Language	Vanilla JavaScript, ES2021-compatible
Interface technology	HTML5 Canvas
External packages	None
Current SHA-256 hash of the implementation file	601d98427e75a058a499f0e269713e10a9c543ceda25c1bc4f0929746dfa6fb5

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